EpisodicMemory · C1

Keep what happened.
Keep disagreement in the record.

ELI16 · THE IDEA IN PLAIN LANGUAGE

A lab notebook: a repeated report does not become a second event, and an inconvenient observation is not erased.

Why it matters: Later belief changes need a durable history of occurrences and provenance. Processing receipts must work even when no numerical belief can be formed.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

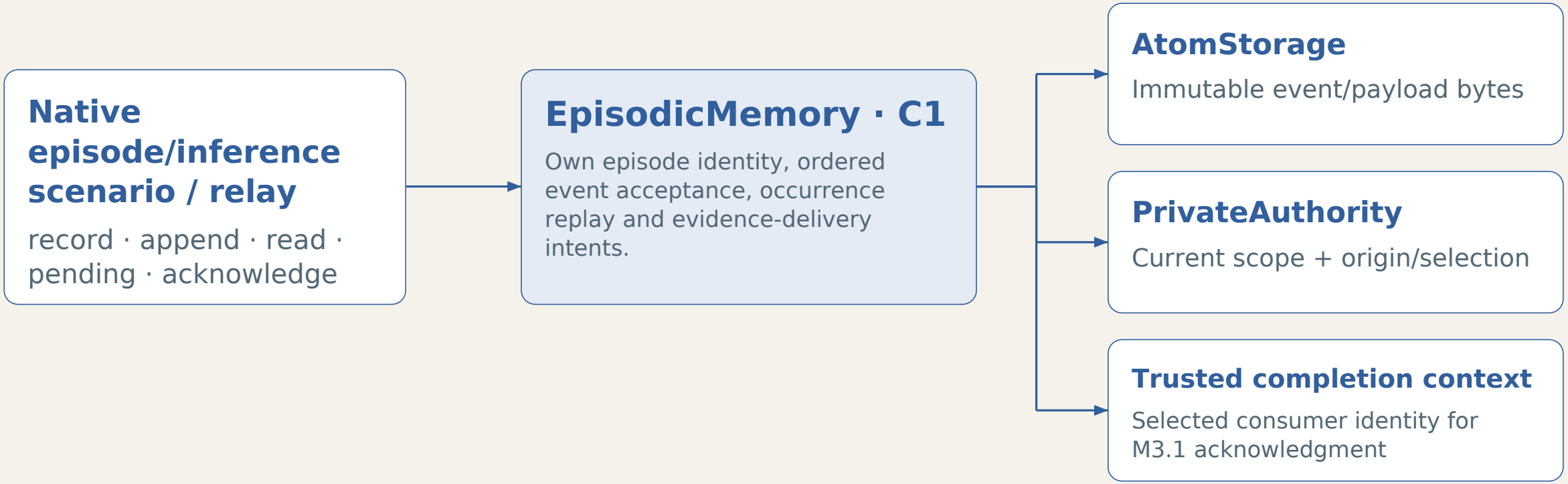
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	Relations	**Episodes · C1**	Messaging
AtomStorage · C0	Search projection	Beliefs · C2	Execution
ConsensusFinality		PLNKernel	Registry
		Attention · C3	Resolver
		Inference · C4	Authorization

Own episode identity, ordered event acceptance, occurrence replay and evidence-delivery intents.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Episode ledger

Identity/head, accepted events, occurrence semantics and replay

Delivery intent

Pending evidence and trusted model-independent completion

OUTSIDE THIS RESPONSIBILITY

Belief and numeric ownership

C2 owns admission; RevisionMath/PLN supply proposals

Foreign owner state

No C2 SQL reads and no interpretation of PLN snapshot internals

episode-local-sequence/2; M3.1 local 3.0.0 adds episode-delivery-completion/1

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
recordEpisode / appendEvent	Create identity; append at expected sequence	Event + delivery intent commit atomically
readEpisode / readEvidence	Read ordered accepted history	Observed time is metadata, not ordering authority
pendingEvidence	Return bounded pending deliveries	Backpressure prevents dropped evidence
acknowledgeDelivery	Accept trusted processing completion	M3.1 local v3; old v2 remains explicit

256 events/episode; 16 pending deliveries; bounded pages; trusted local placement.

A worked example through the boundary

M3.1 REAL DeTTa / native PLN · generated observations; historical source labelled

01 · STEP

Append 16

Accept a full batch of observations.

02 · STEP

Drain 16

Relay validates C2 receipts; C1 clears pending.

03 · STEP

Repeat

Append and drain another sixteen.

04 · STEP

Finish 1

33 retained events; zero pending after each drain.

What the example establishes

Processing completion is not the same as numerical agreement.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

One occurrence

Same occurrence/body replays; altered meaning conflicts.

I2

Ordered acceptance

Expected-head CAS controls append order; observed time cannot retime it.

I3

Atomic intent

An accepted evidence event has a delivery intent in the same C1 commit.

I4

Independent completion

Processing may finish with no numerical truth value; history remains retained.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Payload saved; append not committed	Retry through AtomStorage, then commit episode state.	No cross-owner transaction
C2 decision committed; acknowledgment lost	Relay recovers exact decision and retries trusted completion.	No manufactured acknowledgment
Capacity or malformed receipt	Pause with pending evidence; never drop or falsely complete.	Repeated-drain and hostile-receipt tests

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

33

Observations in one episode

Historical M3.1 repeated-drain observation

16 / 16 / 1

Three drain batches

Pending returned to zero after every batch

256 / 16

Lifetime events / pending limit

Contract bounds; distinct from observed 33

This is a delivery/liveness scenario with generated evidence. No append latency, sustained ingestion, storage growth or long-duration retention benchmark is recorded.

PROPOSED BENCHMARK · NOT RUN

Proposed: vary append/drain ratio, provenance size and injected relay outage; measure p50/p95 append latency, backlog age and bytes/event without dropping history.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	Private SQLite episode owner over real DeTTa; native episode and inference actions.
FIXTURE / LOCAL TEST POLICY	Generated reports, operator origins/grants and hostile completion responses.
BLOCKED / UNSUPPORTED	No required current gate blocked. Pre-existing M3-format pending work is not repaired by new profiles.
UNRUN	Long-lived production histories, distributed writers, deletion/retention policy and migration.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Measure backlog behavior	Rate-controlled append/drain and outages	No loss; bounded backpressure and recovery age
Add explicit retention policy	Provenance and replay obligations first	Deletion cannot silently invalidate accepted evidence
Improve inspection tools	Authorized read-only views with privacy controls	Explain event order and delivery state without exposing identifiers

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

Processing completion is not the same as numerical agreement.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m3_1.py <env> --output <fresh>
```

Fresh explicit closure profiles; full wrapper includes older regressions. Historical 33-event evidence remains tied to its own tested revision.

`fusion/episodes.py`
`fusion/delivery_contracts.py`
`fusion/evidence_relay.py`
`tests/test_m3_1_delivery.py`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.